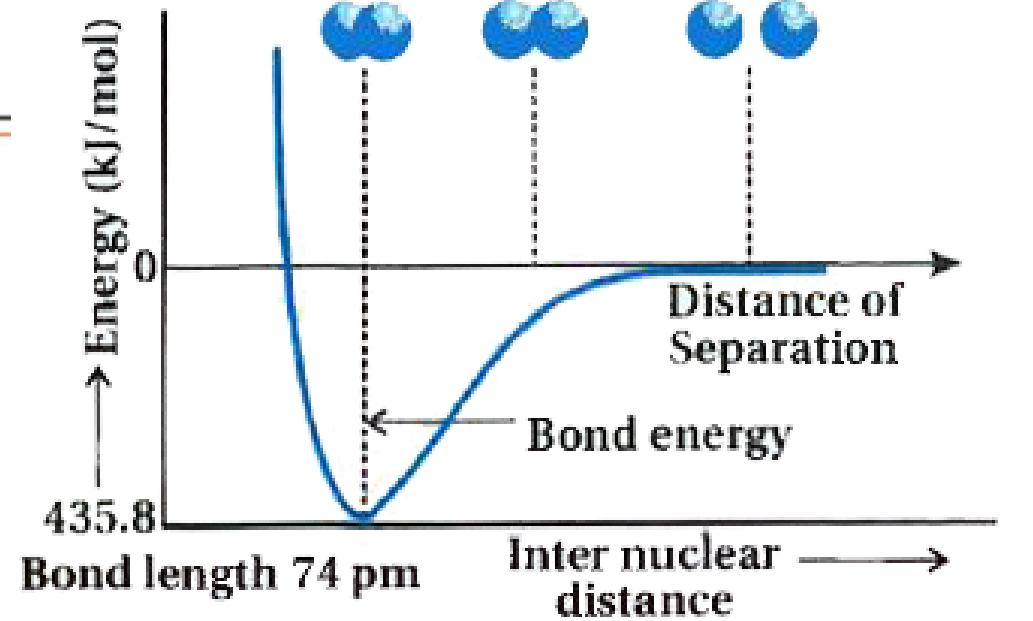
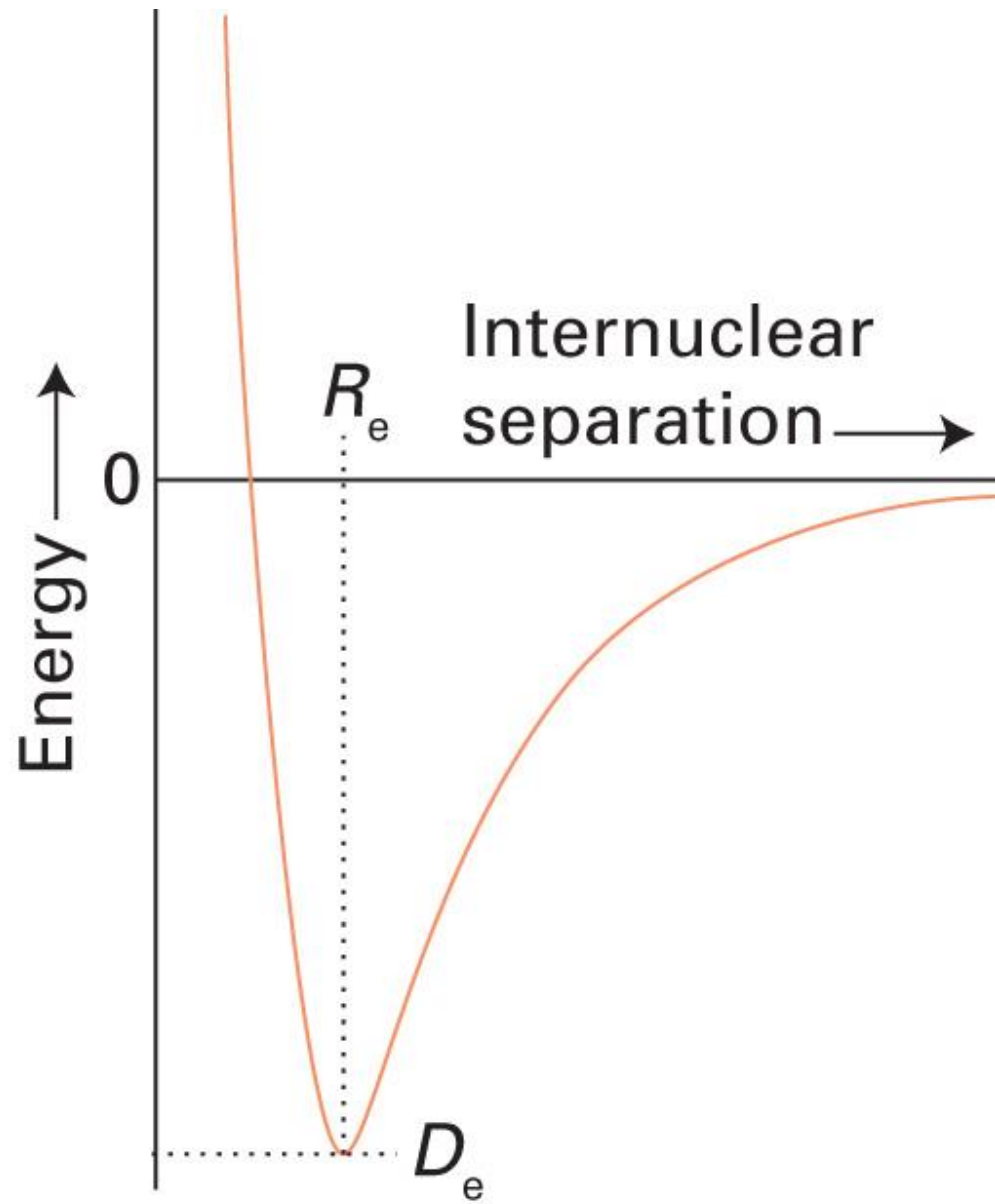
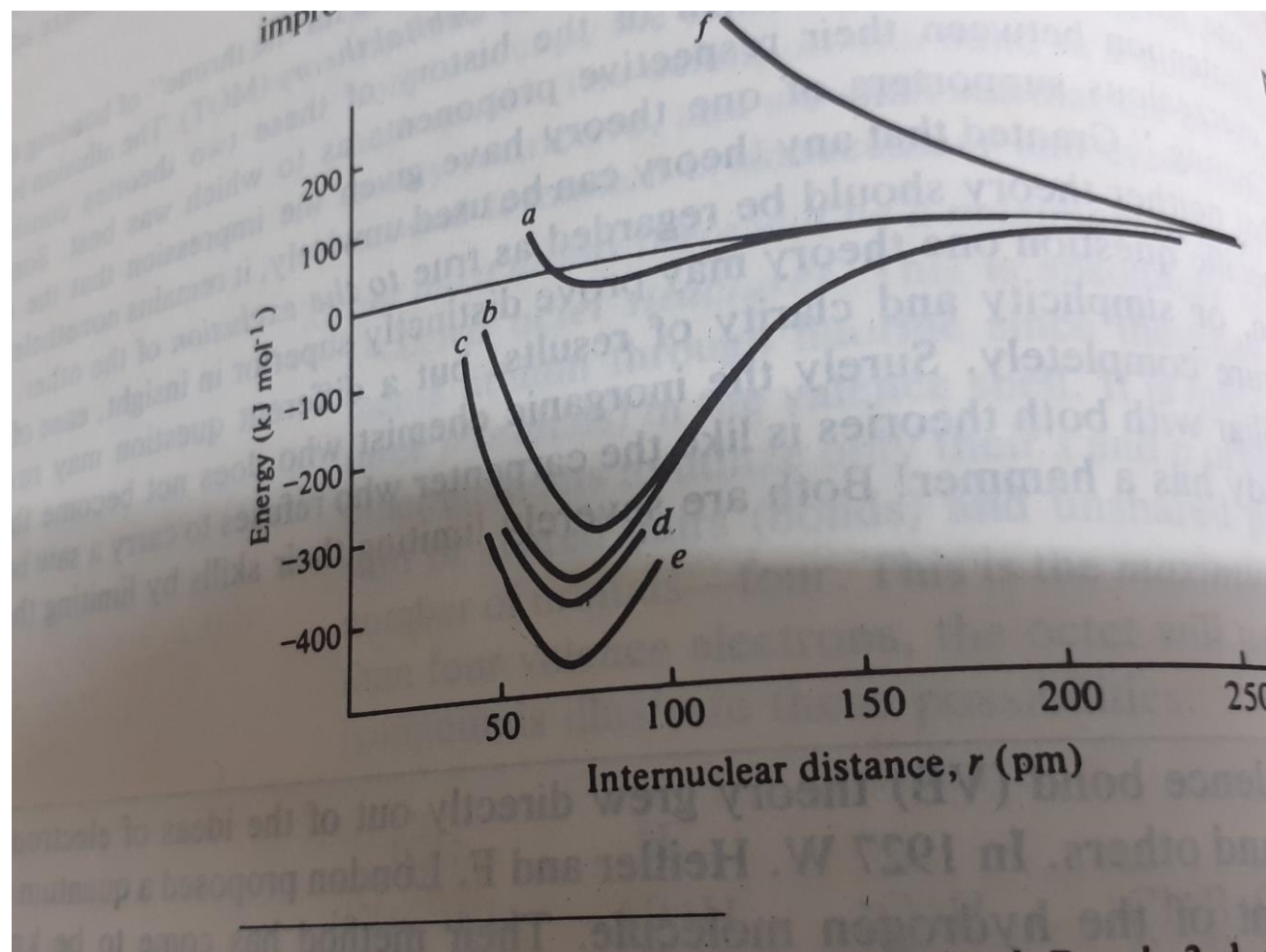


# H<sub>2</sub> molecule



# H<sub>2</sub> Molecule

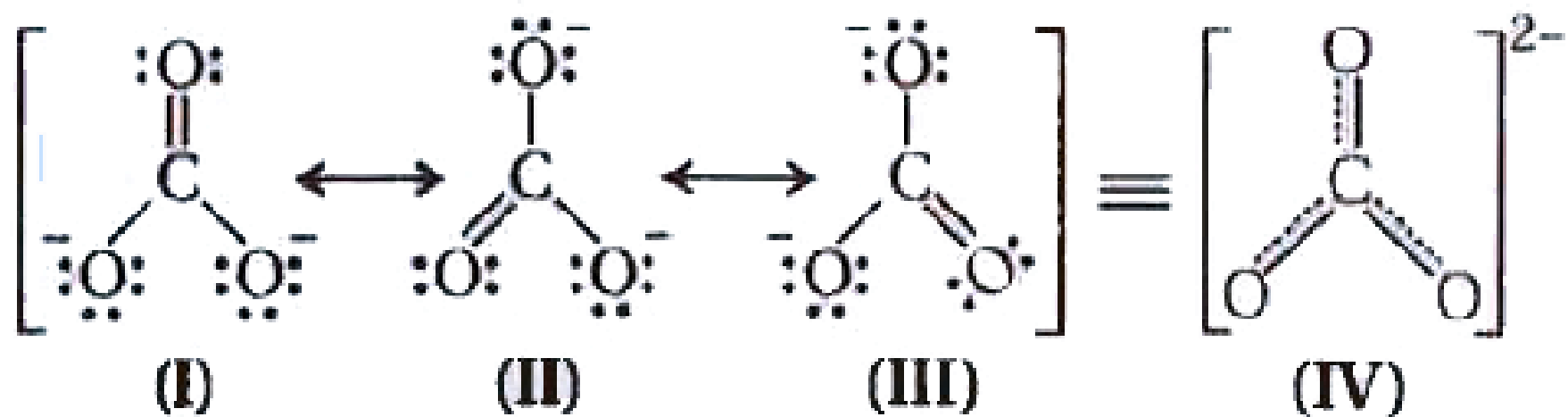


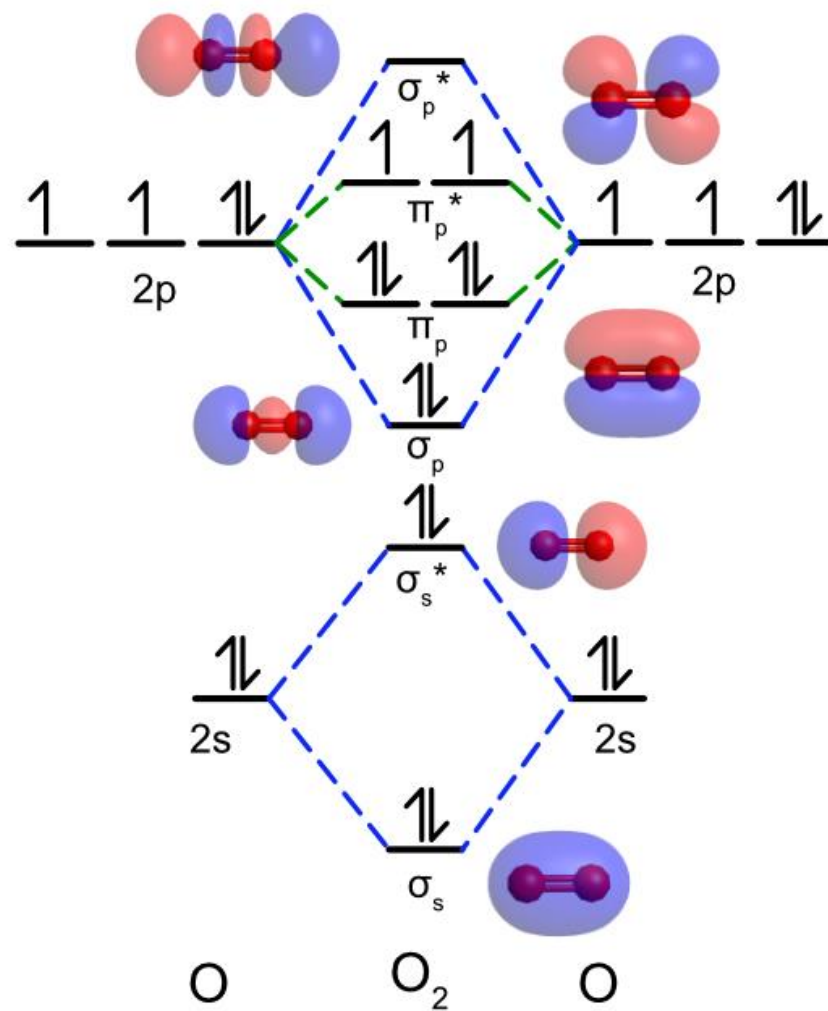
Huheey, 4<sup>th</sup> Ed. Figure 5.1

# Improvements

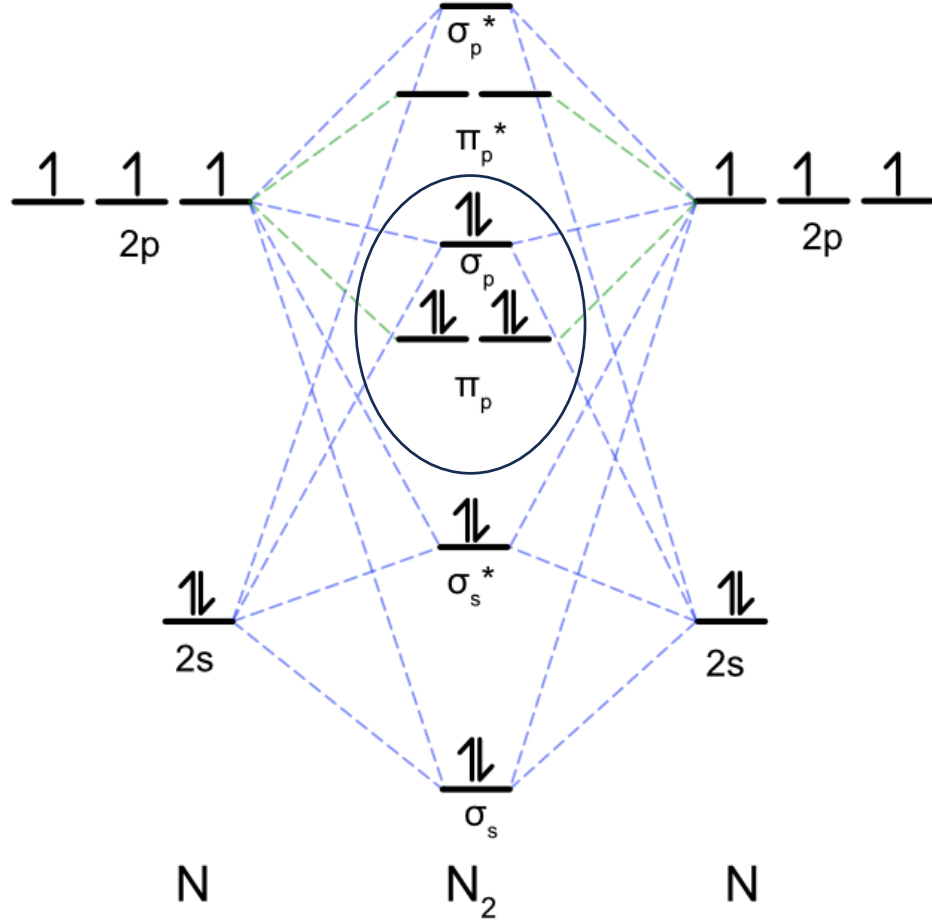
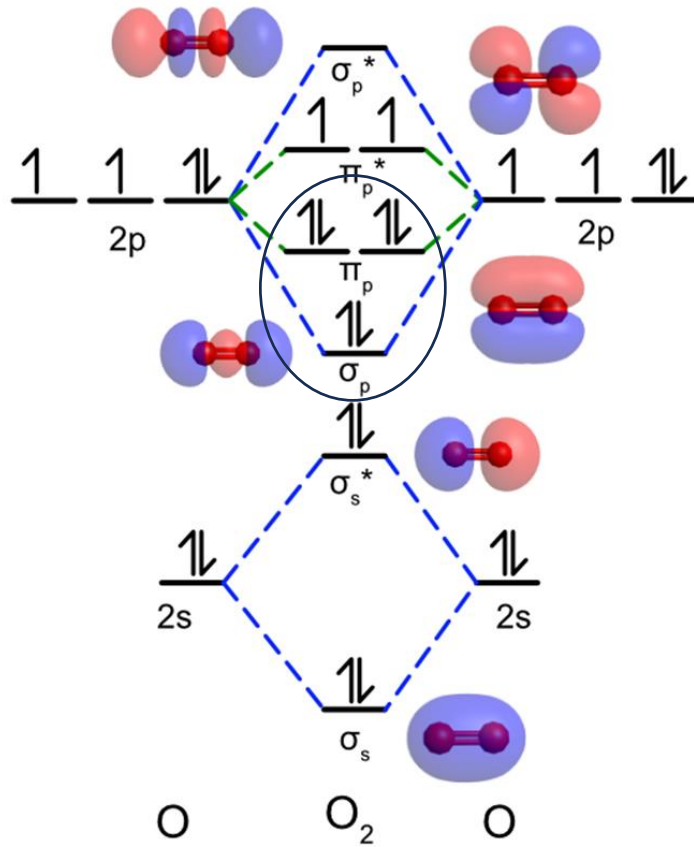
| Type of wave Fubction  | Energy (KJ mol-1) | Distance (pm) |
|------------------------|-------------------|---------------|
| Uncorrected            | 24                | 90            |
| Heitler-London         | 303               | 86.9          |
| Addition of shielding  | 365               | 74.3          |
| Add ionic contribution | 388               | 74.9          |
| Observed value         | 458               | 74.1          |

# Carbonate Resonance

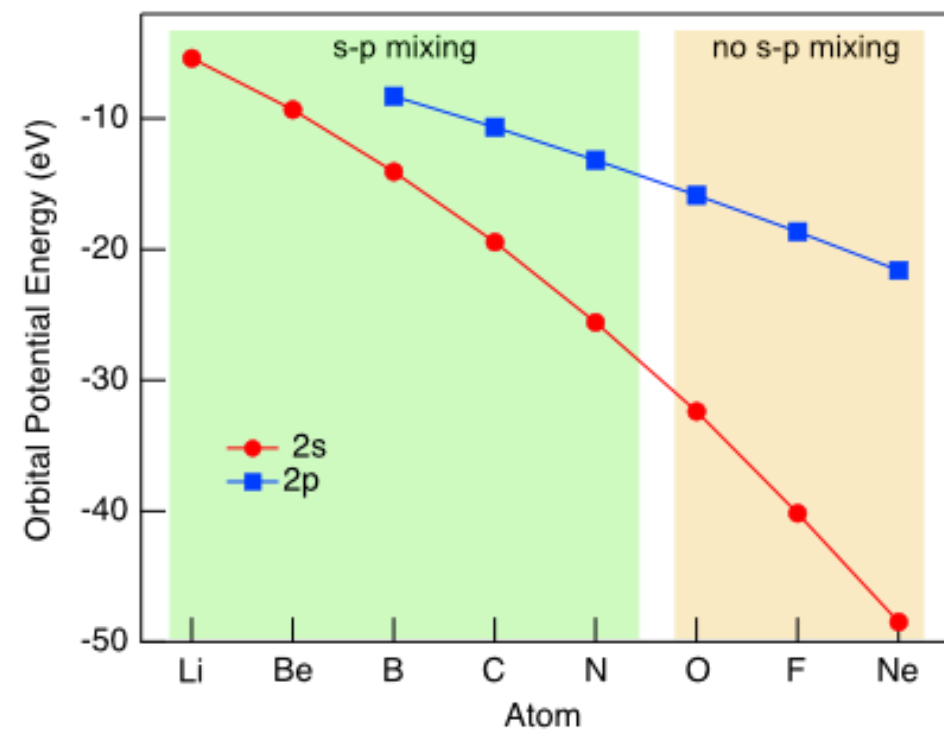


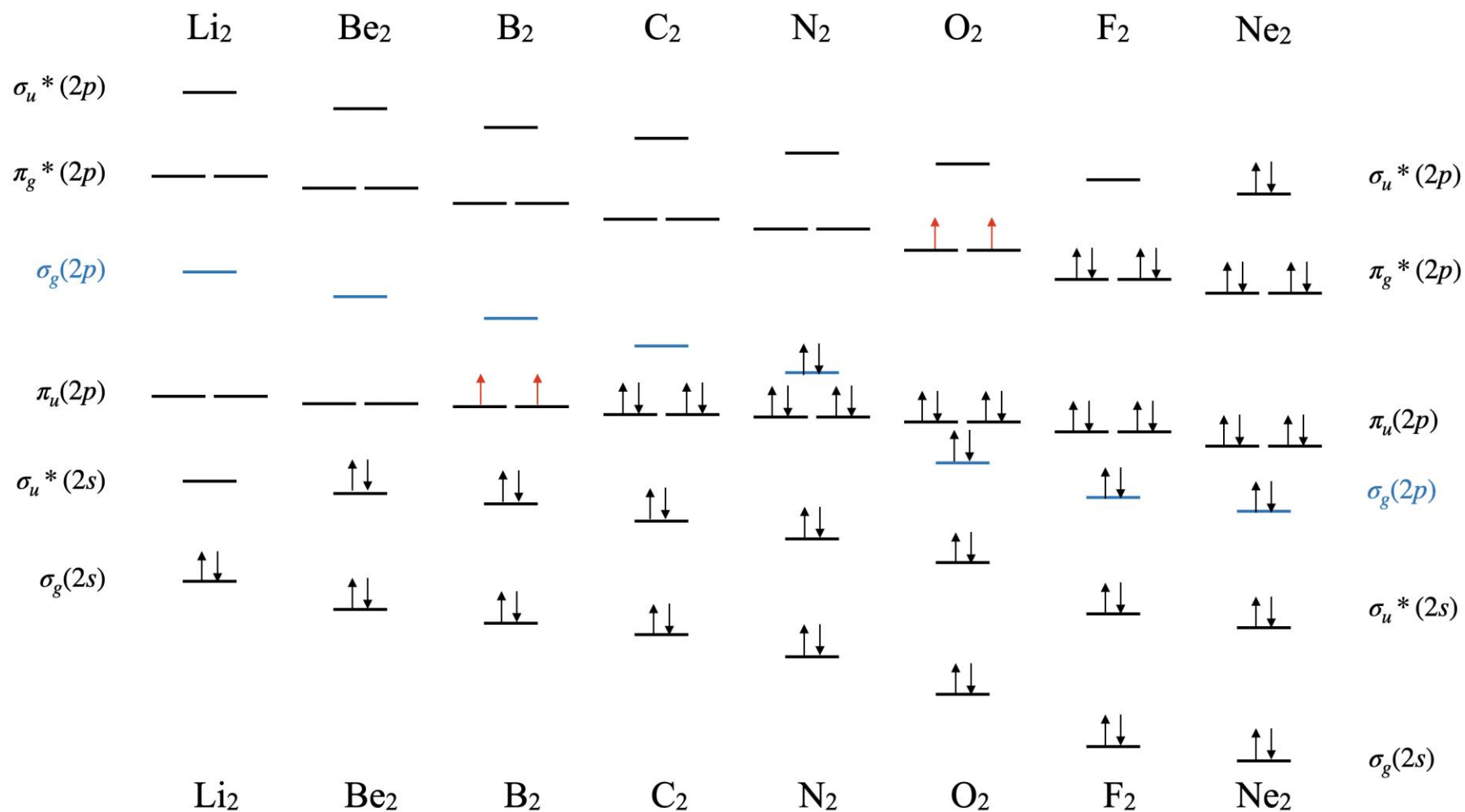


Source: Chem.libretext



Source: Chem.libretext



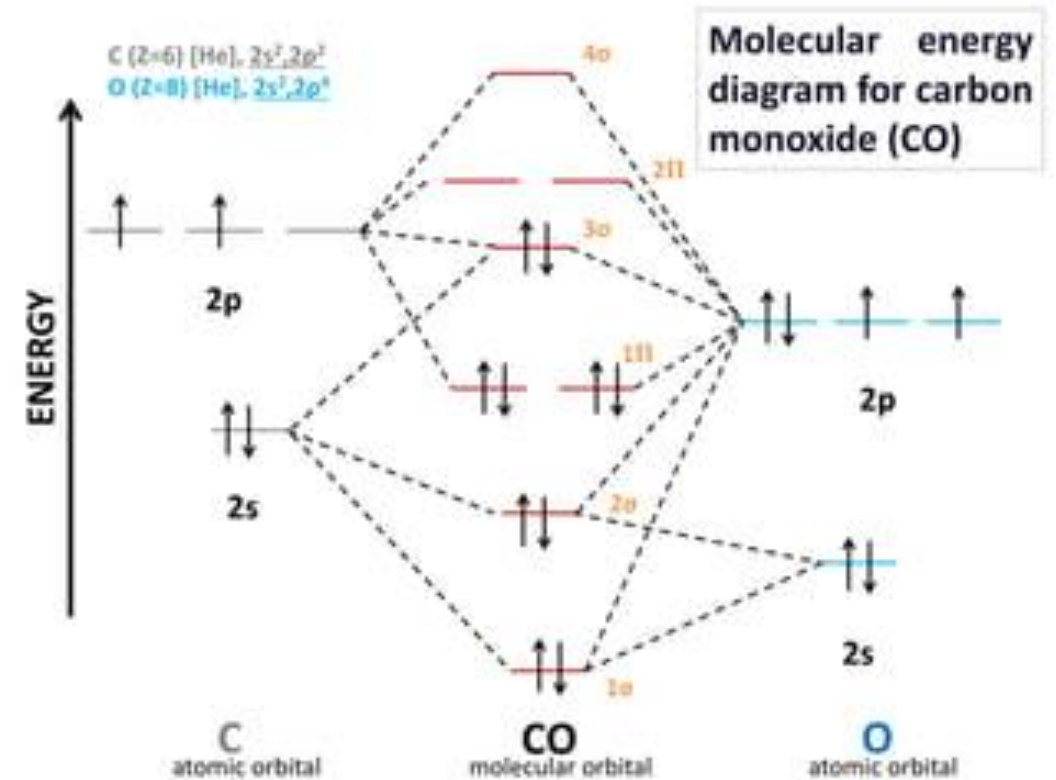
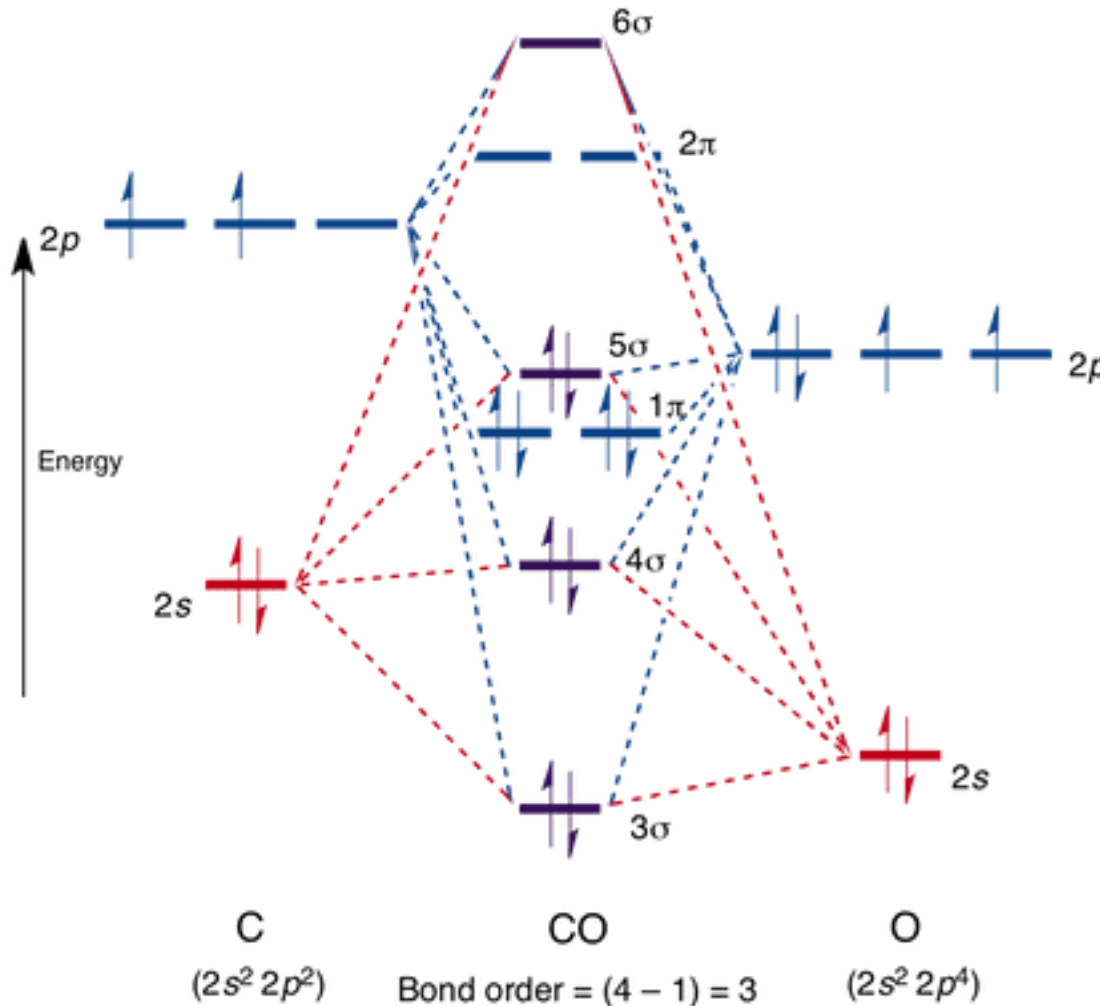


|                                   |       |       |       |       |       |       |       |       |
|-----------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Unpaired e <sup>-</sup>           | 0     | 0     | 2     | 0     | 0     | 2     | 0     | 0     |
| Bond Order                        | 1     | 0     | 1     | 2     | 3     | 2     | 1     | 0     |
| Experimental*<br>Bond Length (pm) | 267.3 | 246.0 | 159.0 | 124.3 | 109.8 | 120.8 | 141.2 | 310.0 |

\*Experimental bond lengths from “List of Experimental Diatomic Bond Lengths”, NIST, <https://cccbdb.nist.gov/diatomicexpbondx.asp>

Source: Chem.libretext

# MO of Carbon Monoxide



Source: Chemtube3D